SWAMI KESHVANAND INSTITUTE OF TECHNOLOGY, MANAGEMENT & GRAMOTHAN, JAIPUR

Department of Computer Science & Engineering — Artificial Intelligence

WEEKLY PROGRESS REPORT (FORM-3)

Oceara MRV — Blockchain-Based Blue Carbon Registry & MRV System

Academic session	2026–27	**Reporting window**	03–10 October 2026
Project type	Final Year Project	**Prepared on**	10 October 2026
Project mentor	Mr. Sushant Singh	**Repository**	b230789yash/oceara-mrv-skit-2026-27

1. Project Overview

Oceara MRV is an academic platform concept for monitoring, reporting and verification of blue-carbon ecosystems. Its technical workstreams combine a web dashboard, remote-sensing and AI/ML workflows, data engineering, and a blockchain-oriented registry for traceable project records.

2. Activity Recorded During the Reporting Window

The public Git history for 03–10 October 2026 records frontend changes merged into the main branch, alongside automated report-archive commits. The table distinguishes individual commits from a merge commit to avoid double-counting.

Date	Contributor / record	Verified repository activity	Evidence
09 Oct 2026	Yuvika Halwai	Improved landowner project-status badges.	7ad2c5d
09 Oct 2026	Yuvika Halwai	Restored the landowner dashboard baseline.	111be9d
10 Oct 2026	Yash Mathur — merge record	Merged pull request #5 for the frontend status-badge improvement. This is a merge commit, not a separate individual feature commit.	cf59bc1
08 & 10 Oct 2026	GitHub Actions	Automated Form-3 PDFs were archived. Bot commits are excluded from individual contributor counts.	Archive

Reporting note: the available public history does not show a separate, non-merge Yash-authored feature commit during this exact weekly window. Earlier committed work is listed below with its original date. Untracked work is not represented as a verified repository commit.

3. Cumulative Verified Project Milestones

These milestones pre-date the reporting window. They are included to show broader project progress, not as work newly completed this week.

Original date	Contributor	Committed milestone	Evidence
30 Sep 2026	Yash Mathur	Implemented the Oceara MRV blockchain registry. The commit includes the Solidity registry contract, ABI/deployment-related files, tests and service integration.	f583d04
20 Sep 2026	Yash Mathur	Added automated Form-3 weekly reporting support.	Repository
Sep–Oct 2026	Vineet	Sentinel-2 retrieval changes and AI/ML module documentation are present in repository history.	Repository
Oct 2026	Mohd. Nomaan	Added project data-dictionary documentation for the data-engineering workstream.	Repository
Sep–Oct 2026	Yuvika Halwai	UI/UX design assets, component documentation and frontend/dashboard changes are recorded in repository history.	Repository

4. Current System Workstreams

Blockchain registry. A smart-contract registry implementation is recorded in Git history with a 30 September 2026 commit date. Local-network configuration and tests should not be described as a public-chain deployment unless separately demonstrated.

Remote sensing and AI/ML. Sentinel-2 retrieval changes and AI/ML documentation are present in the repository. Model performance, carbon estimates and end-to-end validation results should only be added when supported by reproducible tests or evaluation records.

Data engineering. A data dictionary is documented. Further integration should verify field names, data types, provenance and consistency with the application database.

Frontend and dashboard. Landowner dashboard baseline and project-status badge changes were merged into main on 10 October 2026.

Reporting and audit trail. A GitHub Actions workflow archives weekly PDF reports. The existing generator counts non-merge Git commits, so untracked files and unmerged branches are not counted automatically.

5. Next Steps

• Commit and review the pending database migration on its feature branch; it was still untracked in the terminal output shared for this report.
• Confirm the database schema and test the migration in a development environment before applying it to a shared database.
• Run contract tests and record the test output and network used.
• Demonstrate the project-registration flow and its database transaction metadata.
• Validate remote-sensing and AI/ML outputs with reproducible evidence.
• Continue using feature branches and pull requests for reviewable contributions.

6. Repository References

Main repository: github.com/b230789yash/oceara-mrv-skit-2026-27
Report archive: weekly_reports/
Blockchain registry commit: f583d04

Prepared for academic review. This print copy summarizes publicly visible repository evidence and distinguishes current reporting-window activity from earlier milestones. Mentor evaluation and signatures remain to be completed.

Student / Team Lead Yash Mathur Signature: _____	**Project Mentor** Mr. Sushant Singh Signature: _____
Department / Lab Coordinator Signature: _____	**Date** _____